

Problem 1: Compute

$$\oint_C y^2 \, dx + x \, dy$$

where C is the counterclockwise path around the perimeter of the rectangle R with $0 \leq x \leq 2$, $0 \leq y \leq 3$.

Problem 2: Compute

$$\oint_C (y\vec{i} - x\vec{j}) \cdot d\vec{r}$$

where C is the unit circle oriented clockwise.

Problem 3: Compute

$$\oint_C y^3 \, dx + (x^3 + 3xy^2) \, dy$$

where C is the path from $(0, 0)$ to $(1, 1)$ along the graph of $y = x^3$ and from $(1, 1)$ to $(0, 0)$ along the graph of $y = x$.

Problem 4: Compute

$$\oint_C (y^3 \vec{i} + 3xy^2 \vec{j}) \cdot d\vec{r}$$

where C is the unit circle oriented counterclockwise.

Problem 5: Compute

$$\oint_C \left(\arctan(x) + y^2 \right) dx + (e^y - x^2) dy$$

where C is the curve shown below.

